

Clinical Risk Assessment Report: Patient 86

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: High Confidence
- Absolute Predicted Risk: 43.9% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 1.98x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Large Zone Emphasis (CT) (GLZLM_LGZE.CT), which elevated the patient's absolute risk by +7.3%. Biologically, this feature reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.39 [Bottom 4%] (+7.3%)**

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Tumor Pixel Correlation (CT) (GLCM_Correlation.CT) **Value: 1.0 [Top 14%] (+4.5%)**

Measures the linear dependency of gray levels; atypical values corroborate a complex, non-uniform spatial cellular arrangement.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.75 [Avg (72th %ile)] (+2.8%)**

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Patient Age (Age) **Value: 53.0 [Bottom 16%] (+2.2%)**

Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.24 [Bottom 7%] (+2.1%)**

Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Short Run High Gray-Level Emphasis (CT) (GLRLM_SRHGE.CT) **Value: 1.17 [Top 6%] (+2.0%)**

Reflects fragmented, short streaks of high density or metabolism, indicating a disjointed and aggressive tumor architecture.

High Gray-Level Zone Emphasis (CT) (GLZLM_HGZE.CT) **Value: 1.27 [Top 6%] (+1.7%)**

Highlights the distribution of high-density or highly metabolic active regions clustered within the tumor.

Small Zone High Gray-Level Emphasis (PET) (GLZLM_SZHGE.PET) **Value: -0.19 [Avg (65th %ile)] (+1.4%)**

Indicates multiple small areas of intense density or metabolism, a marker of erratic and aggressive cellular proliferation.

3. Protective / Mitigating Factors (Reducing Risk)

No major risk-reducing features observed in the top drivers.

Machine Learning Feature Attribution (SHAP Decision Path)

